KDA-Aware Prefix Caching and Long-Context Scheduling

Deep Dive into Kimi K3 — Chapter 44

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Goal: follow request B from a ten-position text match to a joint restart at 8, a protected private restore, and three emitted tokens — and count each step with an explicit owner.

Roadmap

A joint restart boundary

Lifetime and bytes

Service on one timeline

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- Splicing mixed boundaries fails at once: $(45, 64)$ instead of $(45, 45)$. Multimodal identity adds preprocessing, order and positions — never bypassing the groups.

Lifetime and bytes

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- Retention weighs avoided work per constrained resource; shared pages make independent per-byte sorting non-optimal.

Service on one timeline

Cost, admission and fairness

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All timings are assigned arithmetic, not measurements. Next: Chapter 45's Fleet Scheduling, Admission, and Failure Recovery.